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Robust distributed model predictive platooning control for heterogeneous autonomous surface vehicles☆

Henglai Wei, Qi Sun, Jicheng Chen, Yang Shi *

University of Victoria, Victoria, BC, V8W 3P6, Canada

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ABSTRACT

This paper proposes to apply a robust distributed model predictive platooning control approach for a group of heterogeneous autonomous surface vehicles (ASVs) with the input constraint and bounded external disturbances. The control input for each ASV is composed of two parts: the optimal nominal control input and the ancillary control input. The optimal nominal control input is generated by solving a distributed MPC (DMPC) problem based on the state information of itself and its neighbors. The offline ancillary control law aims to ensure that the actual system state trajectory evolves in a hyper-tube centered along the optimal nominal state trajectory. A coupled inter-vehicle safety constraint is designed for the DMPC optimization problem to guarantee the inter-ASV collision avoidance. Theoretical results on ensuring the feasibility of the proposed robust DMPC algorithm are provided and the closed-loop systems are proved to be input-to-state stable. Numerical simulations are performed to verify the theoretical results.

1. Introduction

The platooning control of autonomous vehicles has attracted an increasing attention in the intelligent transportation area due to its potential to improve road capacity and efficiency (Feng et al., 2019; Guanetti et al., 2018). The platooning coordination of multi-vehicle, which requires vehicles traveling at the desired spacing and speed, further reduces the aerodynamic drag and improves the utilization of the road, especially for heavy-duty vehicles (Alam et al., 2015; Turri et al., 2016). In a similar vein, the platooning control for autonomous surface vehicles (ASVs) shown in Fig. 1 can also hugely improve waterborne transport efficiency and reduce labor costs (Viel et al., 2020). In practice, multi-vehicle systems are subject to various physical constraints, e.g., the control input and state constraints (Zheng et al., 2016), and external disturbances (Lin & Görges, 2020). Therefore, the design of an effective platooning control is one of the primary objectives for ASVs and autonomous vehicles study.

Recently, various notable control methods have been reported in the literature to tackle the autonomous vehicle platoon problem, such as optimal control (Jin & Orosz, 2016), sliding mode control (Wu et al., 2020), delay-based spacing policy (Besselink & Johansson, 2017). In comparison with aforementioned control methods, distributed model predictive control (DMPC) gains attention from platooning control researchers since it is able to explicitly handle hard constraints and optimize the closed-loop performance. Also, DMPC is computationally efficient and applicable to dynamically decoupled vehicle platoon systems (Kianfar et al., 2015; Liu et al., 2018). In Dunbar and Caveney (2012), the authors present a DMPC algorithm for the homogeneous nonlinear vehicle platoon and also derive a sufficient condition to ensure the string stability. In Li et al. (2016), each vehicle can communicate with its predecessor and follower. Based on the local and neighboring state information, the platooning control input for each agent is generated by solving a local DMPC optimization problem with a neighboring γ-gain stability constraint. The string stability constraints described above suppress the spacing error at the price of a decreased feasibility. This may be conservative and result in unsatisfactory control performance. To achieve better control performance, the authors in Zheng et al. (2016) develop a DMPC algorithm without considering the string stability constraint for the heterogeneous autonomous vehicle platoon under four types of unidirectional communication topologies.

However, the methods in Dunbar and Caveney (2012), Li et al. (2016) and Zheng et al. (2016) do not consider external disturbances, which may degrade the control performance, or destroy the stability of overall vehicle systems. Designing a robust controller that can deal with the disturbances is critically important to ensure the platooning task to be successful. In order to tackle this challenging issue, some robust DMPC methods have been developed. Tube-based MPC generally achieves a good tradeoff between the control performance and online computational requirement, and it has been insightfully studied in Mayne et al. (2005) for linear systems and in Mayne et al. (2011) for

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* Corresponding author.
E-mail addresses: henglaiwei@uvic.ca (H. Wei), sunqi@uvic.ca (Q. Sun), jichengc@uvic.ca (J. Chen), yshi@uvic.ca (Y. Shi).

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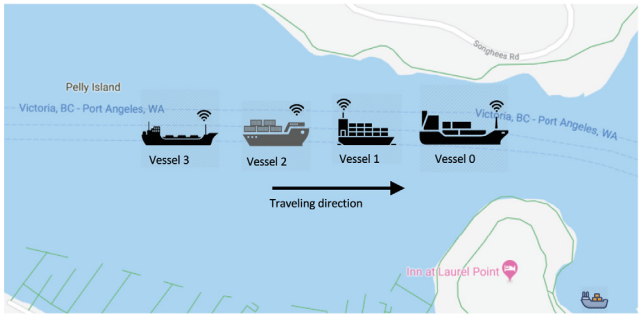


Fig. 1. Illustration of the heterogeneous ASV platoon. The ASVs travel at the desired spacing and speed toward the same direction. ASV 0 is the lead ASV and other ASVs are followers.

nonlinear systems. Further, tube-based decentralized MPC methods for general linear MASs (Trodden & Richards, 2010) and general nonlinear MASs (Nikou & Dimarogonas, 2019) have been developed. A recent paper (Feng et al., 2020) proposes a platooning control algorithm, which combines the tube-based DMPC with event-triggered feedforward control, for homogeneous linear autonomous vehicles. However, the overall system is assumed to be synchronized, which is hard to satisfy. In addition, Feng et al. (2020) does not consider the inter-vehicle safety, while it is necessary in safety-critical environment.

Inspired by the above discussions and existing problems, a nonlinear tube-based DMPC approach for the constrained heterogeneous ASV platoon is proposed. In the proposed algorithm, each ASV is assigned a local controller to calculate its local optimal control input, and then transmits its updated state trajectory to neighbors. The local system states and states of neighbors are required, when each ASV solves the local optimization problem. Since all ASVs optimize control inputs simultaneously, the updated optimal state trajectories cannot be received by its neighbors at the same time instant. Hence, the assumed state trajectory of each ASV, which is used as the optimal state trajectory, will be constructed and exchanged among ASVs. Similar to the methodology used to handle coupled constraints in Richards and How (2007) and Trodden and Richards (2010), the neighbors' state trajectory is also estimated by each local controller. In this way, a coupled constraint is decoupled into a local constraint of the optimization problem. In particular, compared with (Richards & How, 2007; Trodden & Richards, 2010), the transmission of assumed state trajectories among ASVs makes parallel execution of the proposed algorithm a reality.

The main contributions of this paper are summarized as follows:

1. A nonlinear robust DMPC approach is developed for the dynamically decoupled heterogeneous ASV platoon with the input constraint and bounded environmental disturbances. The novelties of the proposed distributed optimization problem lie in that: (a) Different from the method used to guarantee the inter-vehicle collision avoidance in Wei et al. (2019), a coupled minimum safe following distance constraint is designed and incorporated into the optimization problem to ensure the inter-vehicle safety of the ASV platoon; (b) A novel cost term is introduced with the purpose of suppressing the error between the predicted and assumed state of each ASV, which does not require the overall system to be synchronized.
2. The ancillary control law, being designed offline, reduces the deviation between the actual system state and the optimal nominal system state. The ancillary control law design is more computationally efficient in comparison with (Mayne et al., 2011), in which the ancillary control law is determined by solving another MPC problem.
3. Finally, the recursive feasibility of the robust distributed model predictive platooning control algorithm is analyzed, and the heterogeneous ASV platoon is proved to be input-to-state stable.

Table 1
Table of notations for ASV i.

x_i, μ_i	The actual position, and the actual velocity.
$\bar{q}_i, q_i$	The nominal state, and the actual state.
z_i, x_i^e μ_i^e	The actual error state, the actual error position, and the actual error velocity.
$\bar{z}_i, \bar{x}_i^e$ $\bar{\mu}_i^e$	The nominal error state, the nominal error position, and the nominal error velocity.
$\bar{z}_{i,i-1}$	The nominal error state of ASV $i-1$ predicted by ASV i
$\bar{u}_{i,i-1}$	The nominal control of ASV $i-1$ predicted by ASV i
$\bar{u}_i^p, \bar{z}_i^p$	The control input candidate and its corresponding error state.
$\bar{z}_i^*, \bar{u}_i^*$	The optimal nominal error state, and the optimal nominal control input.
π_i, u_i	The ancillary control law, and the actual control input implemented on ASV i.
z_i^a, x_i^a	The assumed error state, and the assumed position.
e_i	The deviation of the actual state from the nominal state.
$\mathbb{U}_i$, $\bar{\mathbb{U}}_i$	The control input constraint set, and the tightened control constraint set.
Ω_i, $\check{o}$	The terminal set, and the bound between the predicted and assumed error state.
d^x d_i^{safe}	The desired spacing between any two ASVs, and the minimum safe following distance.

The remainder of the paper is organized as follows: Section 2 formulates the platooning control problem. In Section 3, the ancillary control law design and the robust DMPC algorithm are presented. The feasibility and stability are proved in Section 4. Section 5 gives the simulation results of the ASV platoon. The conclusion is given in Section 6.

2. System description and control objectives

Let $\mathbb{R}$ and $\mathbb{N}$ denote the sets of real numbers and non-negative integers, respectively. For any vector $x \in \mathbb{R}^n$, $\|x\|$ is the Euclidean norm, $\|x\|_P = x^{\mathrm{T}}Px$ denotes the weighted Euclidean norm, where P is a positive definite matrix. P^{T} denotes the matrix transpose of P. $\lambda_{\min}(P)$ denotes the minimum eigenvalue of the matrix P. Given two sets S_1 and S_2, their Minkowski sum is defined as $S_1 \oplus S_2 := \{s_1 + s_2 | s_1 \in S_1, s_2 \in S_2\}$ and the Pontryagin difference is defined as $S_1 \ominus S_2 := \{s | s + s_2 \in S_1, s_2 \in S_2\}$. A fixed graph, denoted by $\mathcal{G} = (\mathcal{V}^0, \mathcal{E})$, is to describe the information exchange among ASVs, where $\mathcal{V}^0 = \{i \in \mathbb{N} | 0 \le i \le N\}$ is the set of the nodes representing a platoon of ASVs and $\mathcal{E} = \{(i, j) \subset \mathcal{V}^0 \times \mathcal{V}^0\}$ is the set of all directed edges denoting the communication from the ASV i to the ASV j. $\mathcal{N}_i := \{j | (i, j) \in \mathcal{E}\}$ denotes the set of ASV i's neighbors, i.e. ASV i receives the information from ASV $j, j \in \mathcal{N}_i$. $\mathcal{V} = \{i \in \mathbb{N} | 1 \le i \le N\}$ denotes all following ASVs. Some frequently-used notations are shown in Table 1.

2.1. Autonomous surface vehicle modeling

A group of $N+1$ dynamically decoupled nonlinear heterogeneous ASVs are considered in this work. According to Fossen (2002), the model of ASV i is characterized by

$$\dot{\eta}_i = R(\psi_i)\nu_i, \tag{1a}$$

$$M_i\dot{\nu}_i + C_i(\nu_i)\nu_i + D_i(\nu_i)\nu_i = u_i + w_i, \tag{1b}$$

where $\eta_i = [x_i \ y_i \ \psi_i]^{\mathrm{T}}$ includes the surge displacement x_i, sway displacement y_i, and yaw angle ψ_i; the velocity $\nu_i = [\mu_i \ v_i \ r_i]^{\mathrm{T}}$ includes the surge velocity μ_i, sway velocity v_i and yaw velocity r_i. $u_i = [u_{ix} \ u_{iy} \ u_{i\psi}]^{\mathrm{T}}$ is the

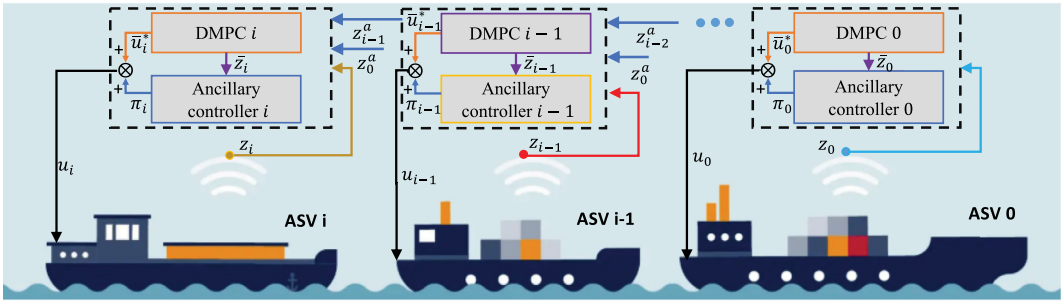


Fig. 2. The control diagram of the proposed control strategy for the ASV platoon[1].

control input and $w_i = [w_{ix}\ w_{iy}\ w_{i\psi}]^{\mathrm{T}}$ denotes the external disturbance. The rotation matrix in yaw $R(\psi_i)$ is

$$R(\psi_i) = \begin{bmatrix} \cos\psi_i & -\sin\psi_i & 0 \\ \sin\psi_i & \cos\psi_i & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

The inertia matrix including the added mass is defined as $M_i = \mathrm{diag}(M_{x_i}, M_{y_i}, M_{\psi_i})$ with $M_{x_i} = m_i - X_{\dot{\mu}_i}$, $M_{y_i} = m_i - Y_{\dot{v}_i}$, and $M_{\psi_i} = I_{z_i} - N_{\dot{r}_i}$. m_i and I_{z_i} are the ship mass and the moment of inertia about the z-axis of the fixed body frame, respectively. The hydrodynamic damping matrix including the drag force and vortex-induced force is

$$\begin{aligned} D_i(v_i) =& \mathrm{diag}(-X_{\mu_i}, -Y_{v_i}, -N_{r_i}) \\ &+ \mathrm{diag}(-X_{\mu_i|\mu_i|}|\mu_i|, -Y_{v_i|v_i|}|v_i|, -N_{r_i|r_i|}|r_i|), \end{aligned}$$

and the Coriolis and centripetal matrix is

$$C_i(v_i) = \begin{bmatrix} 0 & 0 & -M_{y_i} v_i \\ 0 & 0 & M_{x_i}\mu_i \\ M_{y_i} v_i & -M_{x_i}\mu_i & 0 \end{bmatrix}.$$

Due to space limit, detailed definitions of dynamics are referred to Fossen (2002).

2.2. ASV platoon modeling

In this paper, the platooning control of heterogeneous ASVs is studied in the one-dimensional (x-direction) space. Due to the symmetric structure of ASVs, the resulting decoupled surge dynamics of ASV i is given by

$$\dot{x}_i = \mu_i, \tag{2a}$$

$$M_{x_i}\dot{\mu}_i - X_{\mu_i}\mu_i - X_{\mu_i|\mu_i|}|\mu_i|\mu_i = u_{ix} + w_{ix}. \tag{2b}$$

Define $q_i(t) = \left[x_i(t)\ \mu_i(t)\right]^{\mathrm{T}} \in \mathbb{R}^2$ as the actual system state. The predecessor–leader following communication topology proposed in Zheng et al. (2016) is adopted. Under this type of communication topology, ASV i can receive the information from its preceding ASV $i-1$ and the lead ASV 0. Hence, the neighboring set of ASV i is denoted by $\mathcal{N}_i = \{0, i-1\}, i \in \mathcal{V}$. The state of lead ASV 0 is denoted as $q_0(t), t \geq 0$, and the following ASV $i, i \in \mathcal{V}$, will track the desired state,

$$q_i^{\mathrm{des}}(t) = q_0(t) - \begin{bmatrix} id^x \\ 0 \end{bmatrix}, t \geq 0, \tag{3}$$

where d^x is a constant denoting the desired spacing between ASV i and its preceding ASV $i-1$. For ASV $i, i \in \mathcal{V}^0$, the error state $z_i(t) = \left[x_i^e(t)\ \mu_i^e(t)\right]^{\mathrm{T}}$ is given by

$$z_i(t) = q_i(t) - q_i^{\mathrm{des}}(t), \tag{4}$$

where $x_i^e(t)$ and $\mu_i^e(t)$ denote the error position and error velocity, respectively. Taking the time derivative of (4), the error dynamics of ASV i can be described as

$$\begin{aligned} \dot{z}_i(t) =& f_i(z_i(t)) + B_i u_i(t) + B_i w_i(t) \\ =& \begin{bmatrix} \mu_i - \mu_0 \\ M_{x_i}^{-1}(X_{\mu_i}\mu_i + X_{\mu_i|\mu_i|}|\mu_i|\mu_i) - \dot{\mu}_0 \end{bmatrix} \\ &+ B_i u_i(t) + B_i w_i(t), \end{aligned} \tag{5}$$

where $B_i = [0\ M_{x_i}^{-1}]^{\mathrm{T}}$. By a slight abuse of notation, $u_i(t)$ and $w_i(t)$ are used to denote $u_{ix}(t)$ and $w_{ix}(t)$ in the sequel. Here the control input $u_i(t)$ is subject to the control input constraint given by $u_i(t) \in \mathbb{U}_i \subseteq \mathbb{R}$. $w_i(t) \in \mathbb{W}_i := \{|w_i| \leq w_{i,\max}\} \subseteq \mathbb{R}$ represents the additive disturbance.

For the error system in (5), a standard assumption is given as in Dunbar and Caveney (2012).

Assumption 1. For ASV $i, i \in \mathcal{V}^0$, (1) The function $f_i(z_i) : \mathbb{R}^2 \to \mathbb{R}^2$ is continuously differentiable with $f_i(0) = 0$; (2) The system in (5) with any initial state $z_i(0)$ has a unique, absolutely continuous solution for the control input $u_i(t) \in \mathbb{U}_i$ and $w_i(t) \in \mathbb{W}_i$, $t \geq 0$.

Without considering the disturbances $w_i(t)$, the nominal error dynamics of the ASV i can be written as

$$\dot{\bar{z}}_i(t) = f_i(\bar{z}_i(t)) + B_i\bar{u}_i(t), \tag{6}$$

where $\bar{z}_i(t) \in \mathbb{R}^2$ and $\bar{u}_i(t) \in \mathbb{R}$ are the nominal error state and nominal control input, respectively. Next, the definition of input-to-state stable (ISS) is recalled.

Definition 1 (*Input-to-State Stability, Jiang & Wang, 2001*)**.** Consider the system given by $\dot{x}(t) = f(x(t), w(t))$, where $x(t)$ is the system state, and $w(t)$ is the bounded input disturbance. Then, the system is ISS, if there exist a $\mathcal{KL}$ function $\alpha_1(\cdot,\cdot)$ and a $\mathcal{K}$ function $\gamma(\cdot)$, such that,

$$\|x(t)\| \leq \alpha_1(\|x(0)\|, t) + \gamma(\|\boldsymbol{w}\|_{[0,t]}),$$

where $\|\boldsymbol{w}\|_{[0,t]} := \sup\{\|w(s)\| | 0 \leq s \leq t\}$.

2.3. Control objectives

The main control objective of this work is to realize platooning control of a group of heterogeneous ASVs subject to the input constraint and external disturbances. ASV $i, i \in \mathcal{V}$ maintains the desired spacing with its preceding ASV $i-1$ and tracks the speed of the lead ASV 0. Furthermore, in order to ensure the safety of the ASV platoon, the spacing between ASV i and ASV $i-1$ satisfies

$$x_i(t) - x_{i-1}(t) \leq -d_i^{\mathrm{safe}}, \tag{7}$$

where $d_i^{\mathrm{safe}} := \mu_{i,\max} t_s + \mu_{i,\max}^2/2b_s$ is the minimum safe following distance, t_s is the minimum reaction time, b_s is the maximum deceleration, and $\mu_{i,\max}$ is the maximum speed. From the definition of error state in (4), the inequality in (7) can be further written as

$$E(z_i(t) - z_{i-1}(t)) \leq -d_i^{\mathrm{safe}} + d^x, \tag{8}$$

where $E = [1\ 0]$, and $d_i^{\mathrm{safe}} \leq d^x$. It is worth noting that since the predecessor-leader following communication topology is adopted in

[1] Source: https://novimar.eu/concept/.

this paper, ASV i only considers the minimum safe following distance d_i^{safe} between ASV i and its preceding ASV $i-1$.

In this work, a robust DMPC control method is proposed to achieve the control objective of the heterogeneous ASV platoon. As shown in Fig. 2, the control input applied to ASV i consists of two components: (1) the nominal control action $\bar{u}_i^*(t)$, generated by solving the DMPC problem, aims to steer the nominal error state $\bar{z}_i(t)$ toward the origin, and (2) the ancillary control law $\pi_i(z_i(t), \bar{z}_i(t))$, which is designed offline, targets at keeping the actual error state trajectory $z_i(t)$ in a specific set centered along the nominal error state trajectory $\bar{z}_i(t)$.

It is assumed that the DMPC problem for each ASV has the same prediction horizon T and sampling period $\delta \in (0, T]$. Let t_k be the sampling time instant, where $k \in \mathbb{N}$, and $t_{k+1} = t_k + \delta$. At t_k, define $\bar{z}_i(\cdot; t_k)$ as the nominal error state trajectory of ASV i, $\bar{z}_{i,i-1}(\cdot; t_k)$ as the nominal error state trajectory of ASV $i-1$ predicted by ASV i, $\bar{z}_i^*(\cdot; t_k)$ as the optimal nominal error state trajectory, $\bar{z}_{i,i-1}^*(\cdot; t_k)$ as the optimal nominal error state trajectory of ASV $i-1$ predicted by ASV i, $z_i^a(\cdot; t_k)$ as the assumed error state trajectory, and $\bar{z}_i^p(\cdot; t_k)$ as the predicted error state trajectory. The corresponding control inputs are denoted likewise.

As depicted in Fig. 2, ASV $i, i \in \mathcal{V}$ receives the assumed error state trajectory $z_{i-1}^a(s; t_k), s \in [t_k, t_k + T]$ from its preceding ASV $i-1$. The reason of implementing this information transmission strategy is that the optimal nominal error state trajectory $\bar{z}_{i-1}^*(s; t_k), s \in [t_k, t_k + T]$ of the preceding ASV $i-1$ is not available for its following ASV i at the same time instant t_k. Hence, the assumed nominal error state trajectory $z_{i-1}^a(s; t_k), s \in [t_k, t_k + T]$, based on the optimal nominal error states at t_{k-1}, is used to approximate the optimal nominal error state trajectory $\bar{z}_{i-1}^*(s; t_k), s \in [t_k, t_k + T]$. For ASV i, the assumed nominal error state trajectory $z_i^a(\cdot; t_k)$ at time t_k is

$$z_i^a(s; t_k) = \begin{cases} \bar{z}_i^*(s; t_{k-1}), & s \in [t_k, t_{k-1} + T], \\ 0, & s \in (t_{k-1} + T, t_k + T]. \end{cases} \tag{9}$$

Similarly, the assumed control $u_i^a(\cdot; t_k)$ is constructed as

$$u_i^a(s; t_k) = \begin{cases} \bar{u}_i^*(s; t_{k-1}), & s \in [t_k, t_{k-1} + T], \\ 0, & s \in (t_{k-1} + T, t_k + T]. \end{cases}$$

Also, ASV i will receive $z_{i-1}^a(\cdot; t_k)$ of ASV $i-1$.

With these notations, the cost function of ASV $i, i \in \mathcal{V}^0$ at time t_k is defined as

$$\begin{aligned} &J_i\left(\bar{z}_i(t_k), z_{i-1}^a(\cdot; t_k), \bar{u}_i(\cdot; t_k)\right) \\ &= \int_{t_k}^{t_k+T} L_i\left(\bar{z}_i(s; t_k), z_{i-1}^a(s; t_k), \bar{u}_i(s; t_k)\right) \mathrm{d}s \\ &\quad + F_i\left(\bar{z}_i(t_k + T; t_k)\right), \end{aligned}$$

where $\bar{z}_i(t_k) = [\bar{z}_i(t_k)\ \bar{z}_{i,i-1}(t_k)]^{\mathrm{T}}$, $\bar{z}_i(t_k)$ is the initial nominal error state and $\bar{z}_{i,i-1}(t_k) = z_{i-1}^a(t_k; t_k)$ is the initial nominal error state of ASV $i-1$, $\bar{z}_i(s; t_k) = [\bar{z}_i(s; t_k)\ \bar{z}_{i,i-1}(s; t_k)]^{\mathrm{T}}$ and $\bar{u}_i = [\bar{u}_i(s; t_k)\ \bar{u}_{i,i-1}(s; t_k)]^{\mathrm{T}}$. The terminal cost function is defined as $F_i\left(\bar{z}_i(t_k + T; t_k)\right) = \|\bar{z}_i(t_k + T; t_k)\|_{P_i} + \|\bar{z}_{i,i-1}(t_k + T; t_k)\|_{P_{i,i-1}}$ and the stage cost is defined as

$$\begin{aligned} &L_i\left(\bar{z}_i(s; t_k), z_{i-1}^a(s; t_k), \bar{u}_i(s; t_k)\right) \\ &= \|\bar{z}_i(s; t_k)\|_{Q_i} + \|\bar{z}_i(s; t_k) - z_{i-1}^a(s; t_k)\|_{H_i} \\ &\quad + \|\bar{u}_i(s; t_k)\|_{R_i} + \|\bar{z}_i(s; t_k) - z_i^a(s; t_k)\|_{F_i} \\ &\quad + \|\bar{z}_{i,i-1}(s; t_k)\|_{Q_{i,i-1}} + \|\bar{u}_{i,i-1}(s; t_k)\|_{R_{i,i-1}}, \end{aligned} \tag{10}$$

with weighting matrices $P_i \succeq 0$, $P_{i,i-1} \succeq 0$, $Q_i \succeq 0$, $Q_{i,i-1} \succeq 0$, $R_{i,i-1} \succ 0$, $H_i \succeq 0$, $F_i \succeq 0$, $R_i \succ 0$, $H_{-1} = H_0 = Q_{0,-1} = R_{0,-1} = 0$.

Remark 1. The cost term $\|\bar{z}_i(s; t_k) - z_{i-1}^a(s; t_k)\|_{H_i}$ in (10) aims to minimize the predecessor–follower spacing error (i.e., $x_i - x_{i-1} + d^x = z_i - z_{i-1}$), and also maintains the same velocity between itself and its preceding ASV $i-1$. The cost term $\|\bar{z}_i(s; t_k) - z_i^a(s; t_k)\|_{F_i}$ in (10) is introduced for the purpose of suppressing the deviation between the nominal error state $\bar{z}_i(s; t_k)$ and the assumed error state $z_i^a(s; t_k)$. The cost term $\|\bar{z}_{i,i-1}(s; t_k)\|_{Q_{i,i-1}}$ is designed to approximate the error state trajectory of ASV $i-1$.

For the nominal error system in (6), the following assumption of the terminal set is given.

Assumption 2. For the nominal error system of ASV i in (6), there exist an admissible terminal control law $\kappa_i(\cdot) \in \bar{\mathbb{U}}_i$ and a terminal invariant set $\Omega_i(\epsilon_i)$ with a constant $\epsilon_i > 0$, such that: (1) $\forall \bar{z}_i(t_k) \in \Omega_i(\epsilon_i)$, and by implementing $\kappa_i(\cdot)$, $\bar{z}_i(t) \in \Omega_i(\epsilon_i)$ with $t > t_k$; (2) $\forall \bar{z}_i(t) \in \Omega_i(\epsilon_i), i \in \mathcal{V}^0$, $\sum_{i \in \mathcal{V}^0}\left\{\hat{F}_i\left(\bar{z}_i(t)\right) + L_i\left(\bar{z}_i(t), z_{i-1}^a(t), \bar{u}_i(t)\right)\right\} \le 0$.

3. Robust distributed model predictive platooning control

In this section, the DMPC optimization problem and the ancillary control law design are firstly presented. The robust control invariant set is then formulated in Lemma 1. Finally, the robust DMPC algorithm is presented.

3.1. DMPC optimization problem

For ASV $i, i \in \mathcal{V}^0$ at time t_k, the DMPC optimization problem $\mathcal{P}_i$ is designed as

$$\min_{\bar{u}_i(\cdot; t_k)} J_i\left(\bar{z}_i(t_k), z_{i-1}^a(\cdot; t_k), \bar{u}_i(\cdot; t_k)\right)$$

s.t. for $s \in [t_k, t_k + T)$,

$$\bar{z}_i(t_k; t_k) = \bar{z}_i(t_k), \tag{11a}$$

$$\dot{\bar{z}}_i(s; t_k) = f_i(\bar{z}_i(s; t_k)) + B_i \bar{u}_i(s; t_k), \tag{11b}$$

$$\bar{u}_i(s; t_k) \in \bar{\mathbb{U}}_i, \tag{11c}$$

$$\|\bar{z}_i(s; t_k) - z_i^a(s; t_k)\| \le \check{o}, \tag{11d}$$

$$\bar{z}_i(t_k + T; t_k) \in \Omega_i(\epsilon_i), \tag{11e}$$

$$\bar{z}_{i,i-1}(t_k; t_k) = z_{i-1}^a(t_k; t_k), \tag{11f}$$

$$\dot{\bar{z}}_{i,i-1}(s; t_k) = f_{i-1}(\bar{z}_{i,i-1}(s; t_k)) + B_{i-1} \bar{u}_{i,i-1}(s; t_k), \tag{11g}$$

$$\bar{u}_{i,i-1}(s; t_k) = u_{i-1}^a(s; t_k), s \in [t_k, t_{k+1}), \tag{11h}$$

$$\bar{u}_{i,i-1}(s; t_k) \in \bar{\mathbb{U}}_{i-1}, \tag{11i}$$

$$\bar{z}_{i,i-1}(t_k + T; t_k) \in \Omega_{i-1}(\epsilon_{i-1}), \tag{11j}$$

$$E(\bar{z}_i(s; t_k) - \bar{z}_{i,i-1}(s; t_k)) \le d^x - \hat{d}_i^{\text{safe}}, \tag{11k}$$

where the tightened control input constraint $\bar{\mathbb{U}}_i$ is defined as $\bar{\mathbb{U}}_i = \mathbb{U}_i \ominus K_i \mathbb{E}_i \neq \emptyset$, where K_i is the control gain of the ancillary controller and $\mathbb{E}_i$ is the robust control invariant set, and $\hat{d}_i^{\text{safe}} := d_i^{\text{safe}} + e_{i,\max}^x + e_{i-1,\max}^x + \check{o}$. $e_{i,\max}^x$ is the maximum position error between the nominal position and the actual position, which will be given in Section 3.2. Based on the assumed state $z_{i-1}^a(t_k; t_k)$ transmitted from ASV $i-1$, the state trajectory of ASV $i-1$ is locally estimated by ASV i. The terminal constraint is defined as $\Omega_i(\epsilon_i) = \{\bar{z}_i \| \|\bar{z}_i\| \le \epsilon_i\}$, where ϵ_i also satisfies $\epsilon_i < \check{o}$, and $\epsilon_i + \epsilon_{i-1} < d^x - \hat{d}_i^{\text{safe}}$.

Remark 2. The inter-vehicle collision avoidance safety of cooperative autonomous marine vehicles in Wei et al. (2019) is considered by the design of a collision avoidance cost term. In contrast, the inter-vehicle safety of the ASV platoon is guaranteed via a coupled constraint in (11k). The presence of coupled constraints makes the DMPC optimization problem more challenging and interesting in terms of the recursive feasibility analysis. Following the similar approximation strategy in Richards and How (2007) and Trodden and Richards (2013), the nominal error dynamics and constraints of ASV $i-1$ are assumed to be known by the local ASV i in optimization problem $\mathcal{P}_i$. Hence, ASV i is able to approximate the error state trajectory of its preceding ASV $i-1$, which will not be transmitted among ASVs. In this way, the coupled constraint (8) can be decoupled as a local constraint (11k).

Remark 3. The assumed state trajectory $z^a_{i-1}(\cdot;t_k)$ from ASV $i-1$ to ASV i is involved in two aspects: (1) the design of the predecessor-follower spacing error cost term in (10); (2) the initial value in (11f). Because the assumed state trajectory z^a_i in (9) is simultaneously transmitted among ASVs, the nominal control input for each ASV can be determined by solving the local optimization problem in parallel.

3.2. The ancillary controller

The control input that will be applied to ASV i is designed as

$$u_i(s) = \bar{u}^*_i(s;t_k) + \pi_i(z_i(s), \bar{z}^*_i(s;t_k)), s \in [t_k, t_{k+1}). \tag{12}$$

For the sake of simplicity, $\bar{u}^*_i(s)$ is used to denote the optimal nominal control $\bar{u}^*_i(s;t_k)$ and $\pi_i(s)$ is used to the ancillary control law $\pi_i(z_i(s), \bar{z}^*_i(s;t_k))$ in the sequel. The deviation between the actual and nominal error system state can be expressed as

$$e_i(s) = z_i(s) - \bar{z}_i(s;t_k), s \geq t_k, \tag{13}$$

where $e_i(s) = \left[e_{xi}(s)\ e_{\mu i}(s)\right]^{\mathrm{T}}$.

In the next, the robust control invariant set $\mathbb{E}_i$ is derived. The set $\mathbb{E}_i$ centers along the reference state trajectory (i.e., the nominal error state trajectory $\bar{z}_i(s;t_k)$) and the ancillary control law $\pi_i(s)$ ensures that $e_i(s) \in \mathbb{E}_i$.

The following lemma gives the design of the ancillary controller and the robust control invariant set $\mathbb{E}_i$.

Lemma 1. *Suppose that Assumption 1 holds. Let $S_i(e_i(s)) : \mathbb{R}^2 \to [0,\infty)$ be a continuously differentiable function with $0 < S_i(e_i(s)) \leq \alpha_2(\|e_i(s)\|)$, where $\alpha_2(\cdot)$ is a $\mathcal{K}_\infty$ function. Let $\bar{z}_i(s;t_k)$, $s \geq t_k$ be the reference state trajectory. The feedback control law is defined as*

$$\pi_i(s) := -K_i e_i(s), \tag{14}$$

with $K_i = [k_{pi} + k_{di} + d_i,\ M_{x_i} + k_{di}]$, $d_i = -X_{\mu_i} - X_{\mu_i|\mu_i|}|\mu_i|$. If there exist K_i, $\rho_i > 0$, and $\sigma_i > 0$, such that

$$\frac{\mathrm{d}}{\mathrm{d}s} S_i(e_i(s)) + \rho_i S_i(e_i(s)) - \sigma_i w_i^2(s) \leq 0, \forall w_i(s) \in \mathbb{W}_i, \tag{15}$$

where $S_i(e_i(s)) = \frac{1}{2} e_i^T(s) \Gamma_i e_i(s)$, $\Gamma_i = [k_{pi}, M_{x_i}; M_{x_i}, M_{x_i}]$. Then, $\forall e_i(t_k) \in \mathbb{E}_i$, $e_i(s) \in \mathbb{E}_i$, for all $s > t_k$, where

$$\mathbb{E}_i := \left\{ e_i(s) | S_i(e_i(s)) \leq \frac{\sigma_i w^2_{i,\max}}{\rho_i} \right\}. \tag{16}$$

Proof. The proof can be obtained by employing the argument in [Yu et al., 2013, Lemma 1]. Also, it implies that the deviations $|e^x_i(s)|$ and $|e^\mu_i(s)|$ are bounded, i.e., $|e^x_i(s)| \leq e^x_{i,\max}$, $|e^\mu_i(s)| \leq e^\mu_{i,\max}$, with $e^x_{i,\max} > 0$ and $e^\mu_{i,\max} > 0$. □

3.3. Robust DMPC algorithm

The robust DMPC algorithm is implemented in a receding horizon fashion and is presented in **Algorithm 1**.

Algorithm 1 Robust DMPC algorithm

1: **Initialization**: Set $k = 0$. For ASV $i, i \in \mathcal{V}^0$, give the desired state trajectory $q^{\mathrm{des}}_i(\cdot;t_k)$ at t_k. Set $\bar{z}_0(t_k) = 0$ and $\bar{z}_i(s;t_k) = z^a_i(s;t_k) = 0, s \in [t_k, t_k + T]$;
2: ASV 0 samples $z_0(t_k)$, solves $\mathcal{P}_0$ without (11f) – (11k) and broadcasts $z^a_0(\cdot;t_k)$ to ASV $j, j \in \mathcal{V}$;
3: ASV $i, i \in \mathcal{V}$ receives $z^a_\ell(\cdot;t_k), \ell \in \mathcal{N}_i$;
4: ASV $i, i \in \mathcal{V}$ samples $z_i(t_k)$, solves $\mathcal{P}_i$ in (11), and further generates control input trajectory $u_i(s)$ via (12);
5: ASV $i, i \in \mathcal{V}$ applies $u_i(s), s \in [t_k, t_k + \delta)$, and transmits $z^a_i(\cdot;t_k)$ to ASV $i+1$.
6: $k = k + 1$. Go to Step 2.

4. Theoretical analysis

In the following, the theoretical analysis is presented, including the recursive feasibility of the proposed algorithm and the closed-loop robust stability of the heterogeneous ASV platoon with external disturbances.

Theorem 2. *Suppose that Assumptions 1 and 2 hold. For the heterogeneous ASV platoon, if the optimization problem $\mathcal{P}_i$ in (11) is feasible at t_0, then, (1) For ASV $i, i \in \mathcal{V}^0$, problem $\mathcal{P}_i$ is recursively feasible at time $t_k, k \geq 1$ by application of Algorithm 1 if the following conditions hold: $\epsilon_i < \check{o}$, and $\epsilon_i + \epsilon_{i-1} < d^x - \hat{d}^{safe}_i$.*
(2) Furthermore, the actual inter-vehicle safety can be ensured for all admissible disturbances $w_i \in \mathbb{W}_i$.
(3) The overall closed-loop system with the robust DMPC strategy is ISS if the following condition holds: $F_i \geq H_{i+1}$.

Proof. (1) The recursive feasibility of the DMPC algorithm is proved by induction. The optimal nominal control input trajectories obtained at t_k are denoted as $\bar{u}^*_i(s;t_k), \bar{u}^*_{i,i-1}(s;t_k), s \in [t_k, t_k + T)$, which can drive the nominal error systems of ASV i and ASV $i-1$ into the terminal region $\Omega_i(\epsilon_i)$ and $\Omega_{i-1}(\epsilon_{i-1})$, respectively. The optimal nominal error state trajectory with control input trajectory $\bar{u}^*_i(s;t_k), s \in [t_k, t_k + T)$ is given by $\dot{\bar{z}}^*_i(s;t_k) = f_i(\bar{z}^*_i(s;t_k)) + B_i \bar{u}^*_i(s;t_k), s \in [t_k, t_k + T)$. At t_{k+1}, based on the optimal nominal control input trajectory at t_k and Assumption 2, a feasible control input trajectory candidate $\bar{u}^p_i(\cdot;t_{k+1})$ for the problem $\mathcal{P}_i$ is constructed as

$$\bar{u}^p_i(s;t_{k+1}) = \begin{cases} \bar{u}^*_i(s;t_k), & s \in \left[t_{k+1}, t_k + T\right), \\ \kappa_i(\bar{z}_i(s;t_k)), & s \in \left[t_k + T, t_{k+1} + T\right), \end{cases} \tag{17}$$

where $\kappa_i(\cdot)$ is the terminal controller for ASV i. The corresponding state trajectory is computed as

$$\bar{z}^p_i(s;t_{k+1}) = \begin{cases} \bar{z}^*_i(s;t_k), & s \in \left[t_{k+1}, t_k + T\right), \\ z^\kappa_i(s;t_k), & s \in \left[t_k + T, t_{k+1} + T\right). \end{cases} \tag{18}$$

The control input trajectory candidate and the corresponding state trajectory of locally approximated ASV $i-1$ are constructed in a similar way.

According to Assumption 2, $\kappa_i(\bar{z}_i(s;t_k)) \in \bar{\mathbb{U}}_i, s \in [t_k + T, t_{k+1} + T)$. In addition, $\bar{u}^p_i(s;t_{k+1}) = \bar{u}^*_i(s;t_k) \in \bar{\mathbb{U}}_i, s \in [t_{k+1}, t_k + T)$. Hence, the control input constraint (11c) satisfaction is obvious. Further, since the terminal region $\Omega_i(\epsilon_i)$ is invariant with the terminal control law $\kappa_i(\bar{z}_i(s;t_k)), s \in [t_k + T, t_{k+1} + T)$, it has $\bar{z}^p_i(t_{k+1} + T; t_{k+1}) \in \Omega_i(\epsilon_i)$. The terminal state constraint (11e) is satisfied. Due to the assumed state trajectory in (9) and the state trajectory in (18) coincide, i.e., $\bar{z}^p_i(s;t_{k+1}) = z^a_i(s;t_{k+1}), s \in [t_{k+1}, t_k + T]$. In addition, $z^a_i(s;t_k) = 0, s \in (t_k + T, t_{k+1} + T]$ and $\epsilon_i \leq \check{o}$, thus $\|\bar{z}^p_i(s;t_{k+1}) - z^a_i(s;t_{k+1})\| \leq \check{o}, s \in (t_k + T, t_{k+1} + T]$ holds. Hence, the constraint (11d) is fulfilled. The constraints (11i) and (11j) of ASV $i-1$ can be proved following the same logic of (11c) and (11e). Next, for $s \in [t_{k+1}, t_k + T]$,

$$\begin{aligned} &E(\bar{z}^p_i(s;t_{k+1}) - \bar{z}^p_{i,i-1}(s;t_{k+1})) \\ =&E(\bar{z}^*_i(s;t_k) - \bar{z}^*_{i,i-1}(s;t_k)) \\ \leq&d^x - \hat{d}^{\mathrm{safe}}_i. \end{aligned}$$

And for $s \in (t_k + T, t_{k+1} + T]$, since the inequality $\epsilon_i + \epsilon_{i-1} < d^x - \hat{d}^{\mathrm{safe}}_i$ holds, and the terminal constraint implies that (11k) is ensured. Thus, the minimum safe following distance constraint (11k) is satisfied.

(2) At time t_k, from Lemma 1, it can be obtained $z_i(s) \in \bar{z}^*_i(s;t_k) \oplus \mathbb{E}_i$, $s \in [t_k, t_k + \delta]$, $z_{i-1}(s) \in \bar{z}^*_{i-1}(s;t_k) \oplus \mathbb{E}_{i-1}, s \in [t_k, t_k + \delta]$. Since $z^a_{i-1}(t_k;t_k) = \bar{z}_{i,i-1}(t_k;t_k) = \bar{z}_{i-1}(t_k;t_k)$, and $\bar{u}^*_{i,i-1}(s;t_k) = u^a_{i-1}(s;t_k)$, then $\bar{z}_{i,i-1}(s;t_k) = z^a_{i-1}(s;t_k), s \in [t_k, t_k + \delta]$. In addition, by the constraint (11d), it can be obtained $z_{i-1}(s) \in \bar{z}^*_{i,i-1}(s;t_k) \oplus \mathbb{E}_{i-1} \oplus \check{\mathbb{O}}, s \in [t_k, t_k + \delta]$, $\check{\mathbb{O}} := \{\bar{z}_i | \|\bar{z}_i - z^a_i\| \leq \check{o}\}$. Therefore, the actual inter-ASV safety in (8) is ensured since the constraint (11k) holds.

(3) For the overall ASV system, the Lyapunov function is chosen as the sum of the distributed optimal cost function

$$V(t_k) = \sum_{i\in\mathcal{V}^0} J_i\left(\bar{z}_i(t_k), z^a_{i-1}(\cdot;t_k), \bar{u}^*_i(\cdot;t_k)\right). \tag{19}$$

At time t_k, the calculated optimal control input trajectory $\bar{u}^*_i(s;t_k)$, $s\in[t_k, t_k+\delta)$ is applied. At t_{k+1}, applying a suboptimal feasible control input trajectory (17) yields

$$\begin{aligned}
&V(t_{k+1}) = \sum_{i\in\mathcal{V}^0} J_i\left(\bar{z}_i(t_{k+1}), z^a_{i-1}(\cdot;t_{k+1}), \bar{u}^*_i(\cdot;t_{k+1})\right)\\
&\le \sum_{i\in\mathcal{V}^0} J_i\left(\bar{z}_i(t_{k+1}), z^a_{i-1}(\cdot;t_{k+1}), \bar{u}^p_i(\cdot;t_{k+1})\right)\\
&= \sum_{i\in\mathcal{V}^0}\Big\{\int_{t_{k+1}}^{t_{k+1}+T} L_i\left(\bar{z}^p_i(s;t_{k+1}), z^a_{i-1}(s;t_{k+1}), \bar{u}^p_i(s;t_{k+1})\right) \mathrm{d}s\\
&\quad + F_i(\bar{z}^p_i(t_{k+1}+T;t_{k+1}))\Big\}.
\end{aligned}$$

Hence, the difference of the Lyapunov function (19) at t_{k+1} and t_k becomes

$$\begin{aligned}
&V(t_{k+1}) - V(t_k)\\
&\le \sum_{i\in\mathcal{V}^0} J_i\left(\bar{z}_i(t_{k+1}), z^a_{i-1}(\cdot;t_{k+1}), \bar{u}^p_i(\cdot;t_{k+1})\right) - V(t_k)\\
&= \sum_{i\in\mathcal{V}^0} (\Delta^i_1 + \Delta^i_2 + \Delta^i_3),
\end{aligned}$$

where

$$\begin{aligned}
\Delta^i_1 =& -\int_{t_k}^{t_{k+1}} \big\{\|\bar{z}^*_i(s;t_k) - z^a_{i-1}(s;t_k)\|_{H_i}\\
&+ \|\bar{z}^*_i(s;t_k) - z^a_i(s;t_k)\|_{F_i}\\
&+ \|\bar{z}^*_i(s;t_k)\|_{Q_i} + \|\bar{u}^*_i(s;t_k)\|_{R_i}\\
&+ \|\bar{z}^*_{i,i-1}(s;t_k)\|_{Q_{i,i-1}} + \|\bar{u}^*_{i,i-1}(s;t_k)\|_{R_{i,i-1}}\big\}\,\mathrm{d}s,\\
\Delta^i_2 =& \int_{t_{k+1}}^{t_k+T} \big\{L_i\left(\bar{z}^p_i(s;t_{k+1}), z^a_{i-1}(s;t_{k+1}), \bar{u}^p_i(s;t_{k+1})\right)\\
&- L_i\left(\bar{z}^*_i(s;t_k), z^a_{i-1}(s;t_k), \bar{u}^*_i(s;t_k)\right)\big\}\,\mathrm{d}s,\\
\Delta^i_3 =& \int_{t_k+T}^{t_{k+1}+T} \big\{\|\bar{z}^p_i(s;t_{k+1})\|_{Q_i} + \|\bar{u}^p_i(s;t_{k+1})\|_{R_i}\\
&+ \|\bar{u}^p_{i,i-1}(s;t_{k+1})\|_{R_{i,i-1}} + \|\bar{z}^p_{i,i-1}(s;t_{k+1})\|_{Q_{i,i-1}}\\
&+ \|\bar{z}^p_i(s;t_{k+1})\|_{F_i} + \|\bar{z}^p_i(s;t_{k+1})\|_{H_i}\big\}\,\mathrm{d}s\\
&+ \|\bar{z}^p_i(t_{k+1}+T;t_{k+1})\|_{P_i} - \|\bar{z}^*_{i,i-1}(t_k+T;t_k)\|_{P_{i,i-1}}\\
&+ \|\bar{z}^p_{i,i-1}(t_{k+1}+T;t_{k+1})\|_{P_{i,i-1}} - \|\bar{z}^*_i(t_k+T;t_k)\|_{P_i}.
\end{aligned}$$

It is easy to find that, $\Delta^i_1 < 0$ always holds. Since the assumed state trajectory in (9) and the state trajectory in (18) coincide, that is, $\bar{z}^p_i(s;t_{k+1}) = z^a_i(s;t_{k+1}), s\in[t_{k+1}, t_k+T]$. Then, it can be obtained

$$\begin{aligned}
&\|\bar{z}^p_i(t^s_{k+1}) - z^a_i(t^s_{k+1})\|_{F_i} - \|\bar{z}^*_i(t^s_k) - z^a_{i-1}(t^s_k)\|_{H_i}\\
&+ \|\bar{z}^p_i(t^s_{k+1}) - z^a_{i-1}(t^s_{k+1})\|_{H_i} - \|\bar{z}^*_i(t^s_k) - z^a_i(t^s_k)\|_{F_i}\\
=&\|\bar{z}^p_i(t^s_{k+1}) - z^a_{i-1}(t^s_{k+1})\|_{H_i} - \|\bar{z}^*_i(t^s_k) - z^a_i(t^s_k)\|_{F_i}\\
&- \|\bar{z}^*_i(t^s_k) - z^a_{i-1}(t^s_k)\|_{H_i}\\
\le&\|\bar{z}^*_i(t^s_k) - z^a_{i-1}(t^s_k)\|_{H_i} + \|\bar{z}^*_{i-1}(t^s_k) - z^a_{i-1}(t^s_k)\|_{H_i}\\
&- \|\bar{z}^*_i(t^s_k) - z^a_i(t^s_k)\|_{F_i} - \|\bar{z}^*_i(t^s_k) - z^a_{i-1}(t^s_k)\|_{H_i}\\
=&\|\bar{z}^*_{i-1}(t^s_k) - z^a_{i-1}(t^s_k)\|_{H_i} - \|\bar{z}^*_i(t^s_k) - z^a_i(t^s_k)\|_{F_i},
\end{aligned} \tag{20}$$

where (t^s_{k+1}) is used to denote $(s;t_{k+1})$.

Since $H_0 = H_{-1} = 0$ and $F_i \succeq H_{i+1}$, for Δ^i_2,

$$\begin{aligned}
\sum_{i\in\mathcal{V}^0}\Delta^i_2 =& \sum_{i\in\mathcal{V}^0}\Big\{\int_{t_{k+1}}^{t_k+T}\big\{-\|\bar{z}^*_i(s;t_k) - z^a_i(s;t_k)\|_{F_i}\\
&+ \|\bar{z}^p_i(s;t_{k+1}) - z^a_{i-1}(s;t_{k+1})\|_{H_i}\\
&- \|\bar{z}^*_i(s;t_k) - z^a_{i-1}(s;t_k)\|_{H_i}\big\}\,\mathrm{d}s\Big\}\\
\overset{(20)}{\le}& \sum_{i\in\mathcal{V}^0}\Big\{\int_{t_{k+1}}^{t_k+T}\{\|\bar{z}^*_{i-1}(s;t_k) - z^a_{i-1}(s;t_k)\|_{H_i}\\
&- \|\bar{z}^*_i(s;t_k) - z^a_i(s;t_k)\|_{F_i}\}\,\mathrm{d}s\Big\}\\
\le& \int_{t_{k+1}}^{t_k+T} -\|\bar{z}^*_N(s;t_k) - z^a_N(s;t_k)\|_{F_N}\,\mathrm{d}s \le 0.
\end{aligned} \tag{21}$$

With the terminal control law $\kappa_i(\cdot)$ and $\kappa_{i-1}(\cdot)$, the system states will be $\bar{z}^p_i(s;t_{k+1})$ and $\bar{z}^p_{i,i-1}(s;t_{k+1})$, $s\in[t_k+T, t_{k+1}+T]$, respectively. By Assumption 2, it follows that

$$\begin{aligned}
&\sum_{i\in\mathcal{V}^0}\Delta^i_3\\
&= \sum_{i\in\mathcal{V}^0}\Big\{\int_{t_k+T}^{t_{k+1}+T}\{\|\bar{z}^p_i(s;t_{k+1})\|_{Q_i} + \|\bar{u}^p_i(s;t_{k+1})\|_{R_i}\\
&\quad + \|\bar{z}^p_i(s;t_{k+1})\|_{F_i} + \|\bar{z}^p_{i,i-1}(s;t_{k+1})\|_{Q_{i,i-1}}\\
&\quad + \|\bar{z}^p_i(s;t_{k+1})\|_{H_i} + \|\bar{u}^p_{i,i-1}(s;t_{k+1})\|_{R_{i,i-1}}\}\,\mathrm{d}s\\
&\quad + \|\bar{z}^p_i(t_{k+1}+T;t_{k+1})\|_{P_i} - \|\bar{z}^*_{i,i-1}(t_k+T;t_k)\|_{P_{i,i-1}}\\
&\quad + \|\bar{z}^p_{i,i-1}(t_{k+1}+T;t_{k+1})\|_{P_{i,i-1}} - \|\bar{z}^*_i(t_k+T;t_k)\|_{P_i}\Big\}\\
&\le 0.
\end{aligned} \tag{22}$$

According to (20), (21) and (22), it can be obtained

$$V(t_{k+1}) - V(t_k) \le \sum_{i\in\mathcal{V}^0}(\Delta^i_1 + \Delta^i_2 + \Delta^i_3) < 0, \tag{23}$$

which means that the nominal error systems are asymptotically stable. Define $\bar{Z}^*(t) = [\bar{z}^{*\mathrm{T}}_0(t), \dots, \bar{z}^{*\mathrm{T}}_N(t)]^{\mathrm{T}}$, $Z(t) = [z^{\mathrm{T}}_0(t), z^{\mathrm{T}}_1(t), \dots, z^{\mathrm{T}}_N(t)]^{\mathrm{T}}$, $E(t) = [e^{\mathrm{T}}_0(t), e^{\mathrm{T}}_1(t), \dots, e^{\mathrm{T}}_N(t)]^{\mathrm{T}}$, $U(t) = [u_0(t), \dots, u_N(t)]^{\mathrm{T}}$ and $\bar{U}^*(t) = [\bar{u}^{*\mathrm{T}}_0(t), \dots, \bar{u}^{*\mathrm{T}}_N(t)]^{\mathrm{T}}$, $W(t) = [w_0(t), \dots, w_N(t)]^{\mathrm{T}}$. Further, the overall closed-loop actual system and the overall closed-loop nominal system are given by

$$\dot{\bar{Z}}^*(t) = F(\bar{Z}^*(t)) + B\bar{U}^*(t), \tag{24a}$$

$$\dot{Z}(t) = F(Z(t)) + BU(t) + BW(t), \tag{24b}$$

where $F(\cdot) = [(f_0(z_0(t)))^{\mathrm{T}}, \dots, (f_N(z_N(t)))^{\mathrm{T}}]^{\mathrm{T}} \in \mathbb{R}^{2(N+1)\times 1}$ and $B = \mathrm{diag}\{B_0, \dots, B_N\} \in \mathbb{R}^{2(N+1)\times N}$. By [Lemma 4.5 in Khalil, 1996], if the overall nominal error system in (24a) is asymptotically stable, then there exists a $\mathcal{KL}$ function $\alpha_3(\cdot,\cdot)$, such that

$$\|\bar{Z}^*(t)\| \le \alpha_3(\|\bar{Z}^*(t_0)\|, t-t_0), t\ge t_0. \tag{25}$$

By Lemma 1, if $e_i(t_k)\in\mathbb{E}_i$, then $e_i(t)\in\mathbb{E}_i, t\ge t_k$ always holds. The inequality $\frac{1}{2}e_i(t)^{\mathrm{T}}\Gamma_i e_i(t) \le \frac{\sigma_i w^2_{i,\max}}{\rho_i}$ results in $\|e_i(t)\| \le c_i w_{i,\max}$, where $c_i = \sqrt{\frac{2\sigma_i}{\rho_i\lambda_{\min}(\Gamma_i)}}$. By defining a $\mathcal{K}$ function $\gamma_i(\|w_i\|_{[t_0,t]}) = c_i w_{i,\max}$, it can be obtained $\|e_i(t)\| \le \gamma_i(\|w_i\|_{[t_0,t]})$. By noting that if the actual error state $z_i(t)$ satisfies $z_i(t)\in\bar{z}_i(t)\oplus\mathbb{E}_i$ for all admissible disturbance sequences. It implies that $\|z_i(t)\| \le \|\bar{z}_i(t)\| + c_i w_{i,\max}$ holds. Since $z_i(t) = \bar{z}^*_i(t) + e_i(t)$, thus for the overall platoon system it can be obtained that $Z(t) = \bar{Z}^*(t) + E(t)$. By defining a $\mathcal{K}$ function $\gamma(\|W\|_{[t_0,t]}) = c\sqrt{N+1}\|W_{\max}\|$ for the overall closed-loop system, it is easy to obtain $\|E(t)\| \le \sqrt{\sum_{i\in\mathcal{V}^0} c^2_i w^2_{i,\max}} \le \gamma(\|W\|_{[t_0,t]})$, where $c = \max_{i\in\mathcal{V}^0}\{c_i\}$ and $W_{\max} = [w_{0,\max}, \dots, w_{N,\max}]^{\mathrm{T}}$. Therefore, it follows from (25) that

$$\begin{aligned}
\|Z(t)\| &\le \|\bar{Z}^*(t)\| + \|E(t)\|\\
&\overset{(25)}{\le} \alpha_3(\|\bar{Z}^*(t_0)\|, t-t_0) + \gamma(\|W\|_{[t_0,t]}).
\end{aligned} \tag{26}$$

According to Definition 1, the overall closed-loop system in (24b) is proved to be ISS. □

Table 2
Main parameters of the ASVs in the platoon.

ASV i	mass m_i	added mass $X_{\dot{\mu}_i}$	surge linear drag X_{μ_i}	surge quadratic drag $X_{\mu_i\|\mu_i\|}$
0	15.0 kg	−2.00 kg	−0.72 kg/s	−2.05 kg/m
1	18.0 kg	−2.00 kg	−0.80 kg/s	−2.00 kg/m
2	12.0 kg	−3.00 kg	−1.30 kg/s	−1.60 kg/m
3	14.0 kg	−1.60 kg	−1.70 kg/s	−2.30 kg/m
4	13.7 kg	−2.60 kg	−1.35 kg/s	−1.80 kg/m
5	13.0 kg	−1.30 kg	−1.00 kg/s	−1.30 kg/m
6	14.0 kg	−3.60 kg	−1.20 kg/s	−1.30 kg/m

Table 3
Disturbance parameters of the ASV platoon.

i	0	1	2	3	4	5	6
A_i	4.5	5.0	5.5	5.3	6.1	5.89	6.0
b_i	0.5π	0.7π	0.8π	0.75π	0.9π	0.8π	0.65π
c_i	$\pi/2$	0	$\pi/3$	$-\pi/4$	$\pi/5$	$2\pi/3$	$-\pi/6$

The stability analysis is inspired by Dunbar and Caveney (2012), Wang and Ding (2014), which gives the stability proof for the closed-loop system without external disturbances. In this work, the closed-loop stability is proved based on the sum of the distributed cost function at two consecutive time instants t_k and t_{k+1}. However, the stability during the period $[t_k, t_{k+1}], k \in \mathbb{N}$ should also be guaranteed. This will be discussed in future research.

5. Simulation study

In this section, the simulation of the heterogeneous ASV platoon is presented to verify the effectiveness of the proposed robust distributed model predictive platooning control strategy. The simulated multi-ASV platoon consists of seven Cybership II with different configuration parameters (Skjetne et al., 2005), which are selected and shown in Table 2. The control input constraint is $|u_i| \leq 40$N. The desired spacing is chosen as $d^x = 2.5$ m, and the minimum safe following distance is given as $d_i^{\text{safe}} = 0.5$ m.

The speed $\mu_0(t)$ of the desired state trajectory is given by

$$\mu_0(t) = \begin{cases} 1 \text{ m/s}, & 0\text{s} \leq t < 1.6\text{s}, \\ 1 + 1.4t \text{ m/s}, & 1.6\text{s} < t \leq 2\text{s}, \\ 1.56 \text{ m/s}, & 2\text{s} < t \leq 3.9\text{s}, \\ 1.56 - 1.6t \text{ m/s}, & 3.9\text{s} < t \leq 4.4\text{s}, \\ 0.76 \text{ m/s}, & 4.4\text{s} < t \leq 6\text{s}. \end{cases} \tag{27}$$

In the simulation, time-varying disturbances are simulated by $w_i(t) = A_i \sin(b_i t + c_i)$, where A_i, b_i and $c_i, i \in \mathcal{V}^0$ are given in Table 3.

The parameters for the DMPC optimization problem $\mathcal{P}_i$ are designed as follows: The sampling period $\delta = 0.1$s, the prediction horizon $T = 5\delta$. The weighting matrices are chosen as, $Q_i = Q_{i,i-1} = \text{diag}(10^3, 10^2)$, $P_i = P_{i,i-1} = \text{diag}(10^4, 10^3)$, $H_i = \text{diag}(10^2, 10)$, and $F_i = \text{diag}(10^2, 10^2)$. Note that the weighting matrix $R_i = R_{i,i-1} = 10^{-4}$ is chosen so that the control penalty is comparable to the cost terms of Q_i, F_i and H_i. The terminal set level is $\epsilon_i = 0.2$ and the upper bound of the predicted and assumed error state is chosen as $\check{o} = 0.5$.

The robust control invariant set $\mathbb{E}_i$ in (16) is constructed based on the conditions in Lemma 1. In addition, the ancillary controller should be designed to guarantee that the tightened control input constraint set is not empty, i.e. $\bar{\mathbb{U}}_i = \mathbb{U}_i \ominus K_i\mathbb{E}_i \neq \emptyset$. Intuitively, a smaller robust control invariant set leads to a larger admissible control input domain of the DMPC problem $\mathcal{P}_i$. The control gains k_{pi}, k_{di} and the variables σ_i and ρ_i for ASV $i, i \in \mathcal{V}^0$ are shown in Table 4. Then, the tightened control input constraint sets are: $|\bar{u}_0| \leq 20.67$N, $|\bar{u}_1| \leq 22.16$N, $|\bar{u}_2| \leq 21.32$N, $|\bar{u}_3| \leq 22.93$N, $|\bar{u}_4| \leq 26.24$N, $|\bar{u}_5| \leq 26.02$N, $|\bar{u}_6| \leq 21.38$N.

The simulation results of the heterogeneous ASV platoon using the proposed robust DMPC strategy are demonstrated in Figs. 3–5. Using the proposed strategy yields the position trajectory which demonstrates a good platooning control performance as shown in Fig. 3. The dash line represents the nominal position of the ASV platoon, the solid line represents the actual position, and the shading area represents a hyper-tube whose center is the nominal state trajectory. As shown in Fig. 4, the error $e_i(t)$ between the nominal error state $\bar{z}_i(t)$ and the actual error state $z_i(t)$ stays within a small bounded set and the nominal error state $\bar{z}_i(t)$ approximately converges to zero. Fig. 5 shows the control input trajectories of seven heterogeneous ASVs. It can be observed that the control input constraint of each ASV can be always satisfied.

Table 4
The ancillary control gain K_i of the ASVs.

ASV i	0	1	2	3	4	5	6
k_{pi}	6.0	7.0	9.0	4.0	9.0	10.0	8.0
k_{di}	6.0	8.0	6.0	4.0	8.0	8.0	7.0
ρ_i	0.92	0.85	1.32	1.13	0.75	1.0	0.76
σ_i	0.20	0.10	0.25	0.15	0.05	0.08	0.09

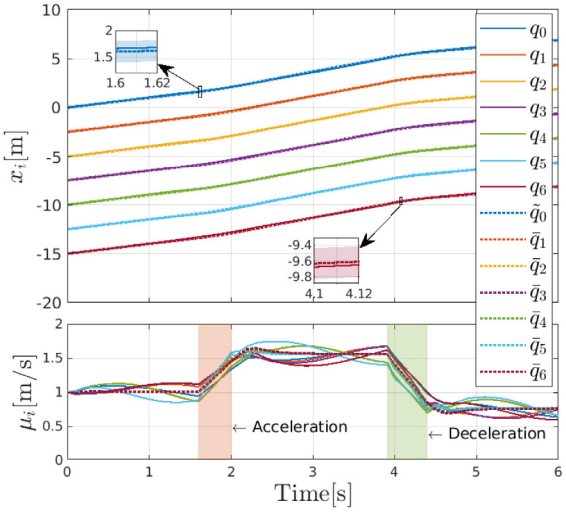


Fig. 3. State trajectory q_i and $\bar{q}_i$ of the ASVs by using the proposed robust DMPC.

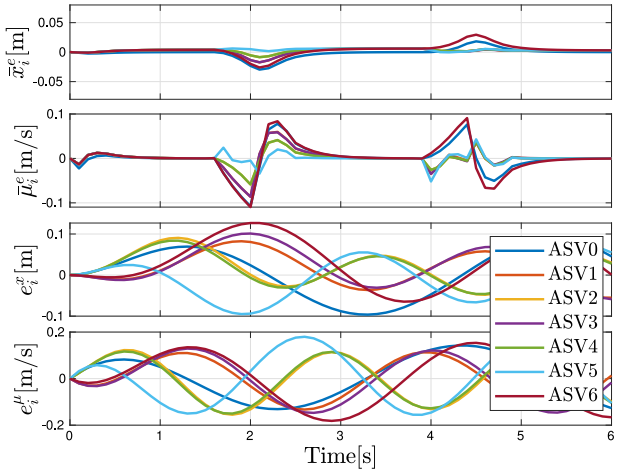


Fig. 4. Trajectories of $\bar{z}_i$ and e_i of the ASVs by using the proposed robust DMPC.

6. Conclusion

In this paper, a robust DMPC approach is designed for the heterogeneous ASV platoon with external disturbances over the fixed

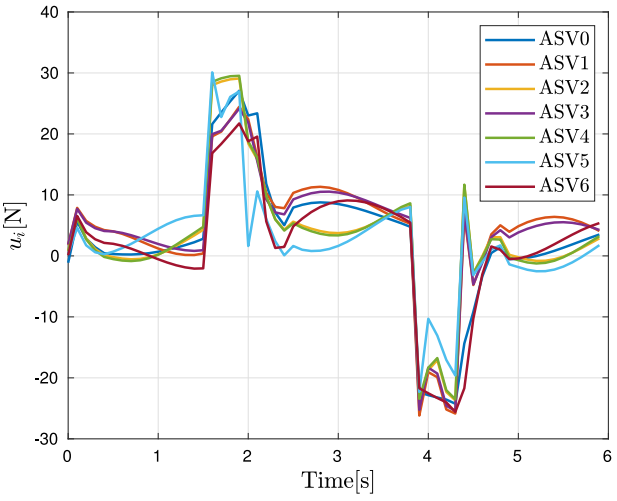


Fig. 5. Control input u_i of the ASVs by using the proposed robust DMPC.

predecessor–leader following communication topology. Each ASV receives the assumed state trajectory from its predecessor and the lead ASV, samples its states and broadcasts its predicted state trajectory to its following ASV. Further, by incorporating the coupled minimum safe following distance constraint into the DMPC optimization problem, the local safe control input is generated for each ASV. The recursive feasibility and closed-loop stability of the proposed method are rigorously analyzed. Simulation studies verify the effectiveness of the proposed distributed platooning control method. Future research can be focused on the design of resilient DMPC for the ASV platoon under cyber attacks, such as DoS attacks (Sun et al., 2019). Another interesting extension of this paper is to investigate the ASV platoon with unknown model parameters using the adaptive MPC approach (Zhang & Shi, 2020).

CRediT authorship contribution statement

Henglai Wei: Data curation, Writing - original draft. **Qi Sun:** Data curation, Writing. **Jicheng Chen:** Data curation, Writing. **Yang Shi:** Writing - original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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